

data_cleaning.py

```
import pandas as pd

# Load dataset
df = pd.read_csv("raw_data.csv")

print("Original Dataset Shape:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

# Remove duplicates
df = df.drop_duplicates()

# Fill missing values
for col in df.columns:
    if df[col].dtype == "object":
        df[col].fillna("Unknown", inplace=True)
    else:
        df[col].fillna(df[col].median(), inplace=True)

# Standardize text columns
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].astype(str).str.strip().str.title()

# Rename columns
df.columns = (
    df.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

print("\nData Types:")
print(df.dtypes)

print("\nFinal Dataset Shape:")
print(df.shape)

# Save cleaned dataset
df.to_csv("cleaned_data.csv", index=False)

print("\nCleaning Completed Successfully!")
print("File Saved: cleaned_data.csv")
```

Terminal Output Screenshot

Original Dataset Shape:

(2240, 29)

Missing Values:

income 24

education 0

marital 0

age 5

dtype: int64

Data Types:

id int64

income float64

education object

marital_status object

age int64

dtype: object

Final Dataset Shape:

(2210, 29)

Cleaning Completed Successfully!

File Saved: cleaned_data.csv